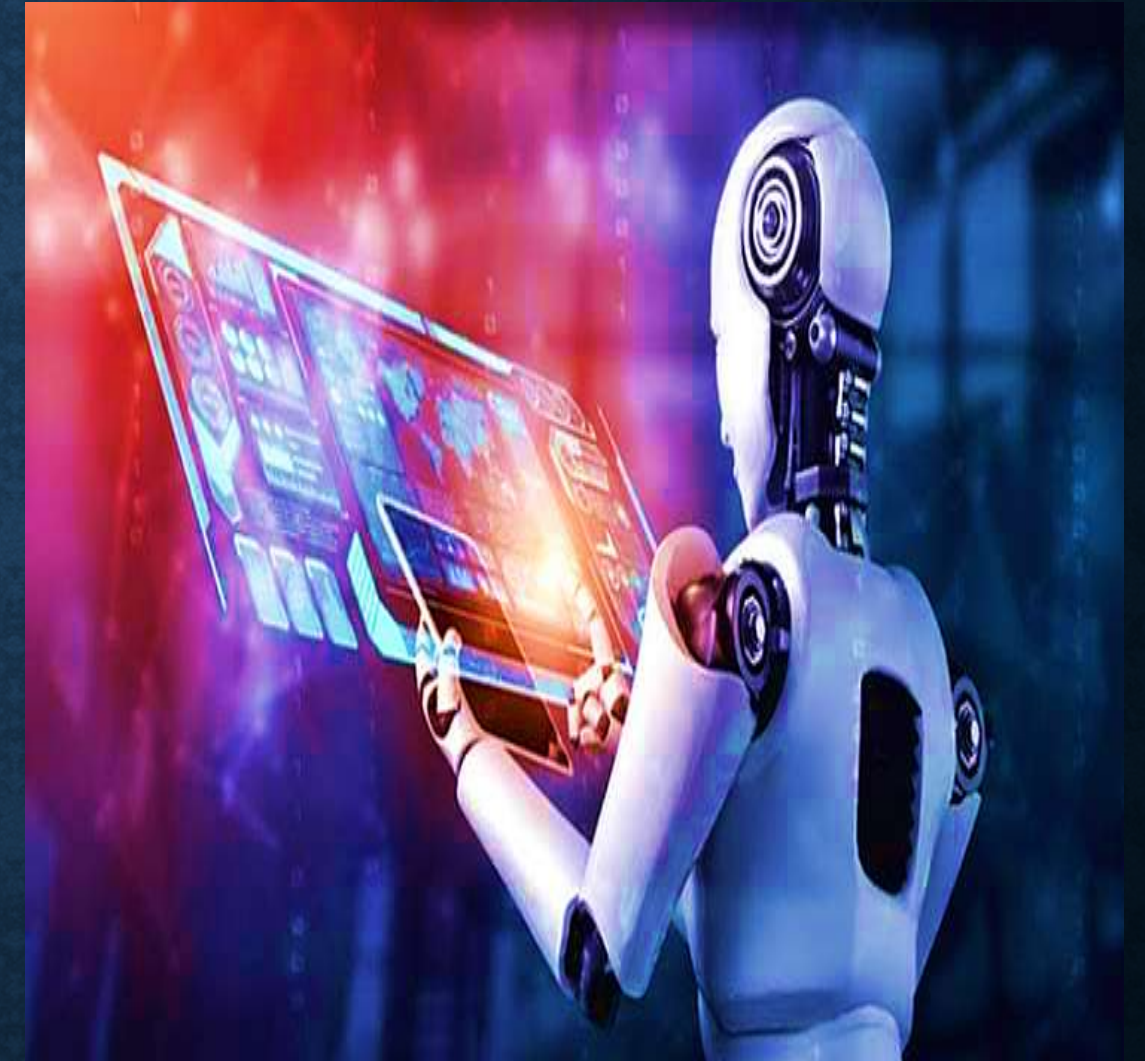


ENGINEERING EXPLORATION



WHAT IS ENGINEERING?

- ❖ Engineering is the practice of applying scientific and mathematical principles to design, build, and optimize structures, machines, and processes.
- ❖ It solves real-world problems by balancing constraints like cost, safety, and efficiency to improve human life.



DIFFERENCE BETWEEN SCIENCE AND ENGINEERING



SCIENCE VS ENGINEERING

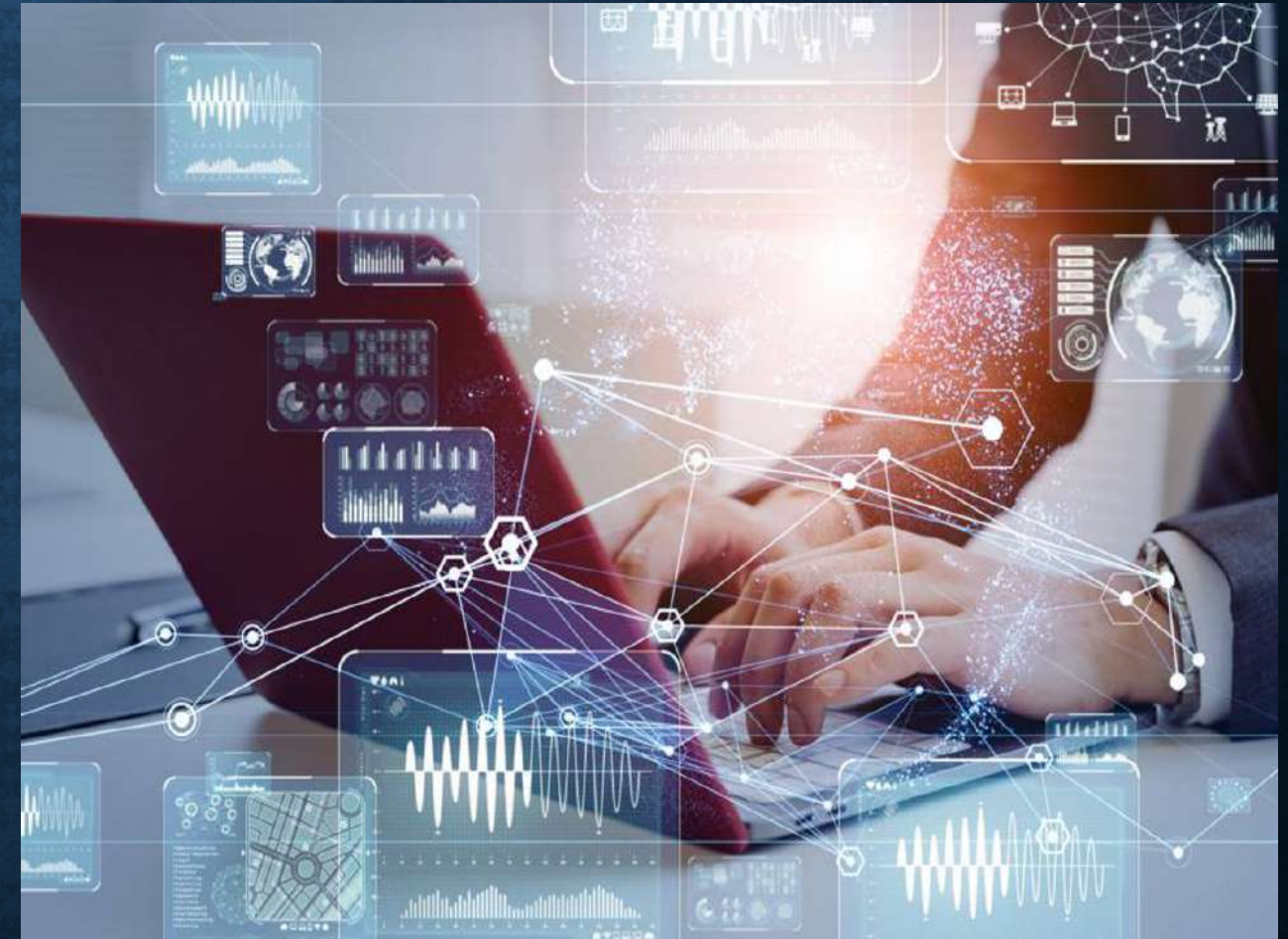
- *SCIENCE*

- ✓ Discover knowledge
- ✓ Studies of natural world
- ✓ EX: studying gravity OR electricity

- *ENGINEERING*

- ✓ Create solution
- ✓ Engineering applies scientific knowledge to solve problems
- ✓ EX: designing bridges, computers, OR mobile apps

DIFFERENCE BETWEEN SCIENTIST AND ENGINEER



SCIENTIST

- ❖ **Scientists** focus on discovering how the natural world works by formulating theories and conducting experiments to expand human knowledge.

ENGINEERING

- ❖ **Engineers** apply existing scientific and mathematical principles to design, build, and optimize practical products, structures, and systems that solve specific real-world problems.
[\[1\]](#), [\[2\]](#), [\[3\]](#)

Needs

- ☐ Strong programming skills (Python, Java, C++, JavaScript, etc.)
- ☐ Knowledge of data structures and algorithms
- ☐ Understanding of AI, cloud computing, cybersecurity, and databases
- ☐ A reliable computer and high-speed internet
- ☐ Problem-solving and analytical thinking

Wants

- ☐ High-paying job
- ☐ Flexible or remote work
- ☐ Opportunities to work on cutting-edge technologies
- ☐ Modern gadgets and advanced development tools
- ☐ Career growth and promotions

DISCIPLINES OF ENGINEERING



Mechanical Engineering:

Vehicle design, engine, brakes Develops control software and robotics algorithms

Electrical Engineering:

Battery and power management Creates battery monitoring software

Electronics and communication Engineering:

Sensors, cameras, communication modules Develops sensor data processing and IoT applications

Civil engineering:

Road infrastructure and traffic systems Creates navigation and smart traffic software

Biomedical Engineering:

Driver health monitoring sensors Develops health monitoring applications

Smart Car

One big product touching multiple engineering department at once

Department Involved	CSE's Contribution
Mechanical Engineering	Engine control software, chassis simulation, braking systems
Electrical Engineering	Battery Management System (BMS), power distribution, EV charging software
Electronics & Communication	Sensors, cameras, GPS, Bluetooth, Wi-Fi, V2X communication
Automobile Engineering	Self-driving (Autonomous Driving), infotainment system, cruise control
Civil Engineering	Smart traffic management, road mapping, EV charging infrastructure

Key Features of a Smart Car

-  Autonomous (Self-driving) Technology
-  GPS Navigation
-  Mobile App Control
-  Smart Battery Management
-  360° Camera & Parking Assist
-  Automatic Emergency Braking
-  Cloud-based Software Updates

Myth ❌ vs Reality ✅

- Only Coding ❌ → Many technologies (AI, Cloud, Cybersecurity) ✅
- Only Software Jobs ❌ → Many Career Options ✅
- Need to be a Genius ❌ → Practice & Interest Matter ✅
- Works Alone ❌ → Teamwork is Important ✅
- Only Computer Work ❌ → Solves Real-World Problems ✅
- AI Will Replace CSE ❌ → AI Helps CSE Engineers ✅

Bottom Line

Skills Required for a Computer Science Engineer in the 21st Century

Technical Skills

- AI & Machine Learning
- Data Structure
- Cloud Computing
- Cybersecurity
- Problem Solving
- programming

Soft Skills 🤝

- Critical Thinking
- Creativity
- Adaptability
- Time Management
- Teamwork

Thank You

The image features a dark blue background with a complex network of glowing blue lines that resemble a circuit board or data pathways. These lines are interspersed with soft, out-of-focus light spots in shades of blue and orange, creating a bokeh effect. The text "Thank You" is prominently displayed in the center, rendered in a bright blue, three-dimensional font with a glowing outline. A faint, semi-transparent watermark is visible across the middle of the image.